Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2025

Course: B.Tech | ***Time: 03 Hours***

Code: 104504 | ***Subject: Linear Integrated Circuits and Applications*** | ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.

(ii) There are NINE questions in this paper.

(iii) Attempt FIVE questions in all.

(iv) Question No. 1 is compulsory.

Q.1 **Choose the correct option / answer the following (Any seven question only):** **[2 x 7 = 14]**

(a) In a monolithic IC fabrication process, boron diffusion is carried out for 2 hours at 1100°C. If the diffusion coefficient increases exponentially with temperature, which fabrication parameter primarily controls junction depth?

(i) Oxide capacitance
(ii) Diffusion time and temperature
(iii) Metal interconnect thickness
(iv) Crystal orientation only

(b) An ideal operational amplifier with open loop gain 10^6 is connected in a non-inverting configuration having $R_f = 90k\Omega$ and $R_1 = 10k\Omega$. The closed loop gain is closest to

(i) 1
(ii) 9
(iii) 10
(iv) 100

(c) A practical differentiator using OP-AMP becomes unstable at high frequencies mainly because of

(i) Excessive gain for high frequency noise
(ii) Infinite slew rate
(iii) Very low input impedance
(iv) Infinite bandwidth

(d) In an instrumentation amplifier, the major purpose of using three OP-AMPs is to obtain

(i) Low bandwidth
(ii) High output resistance
(iii) Low slew rate
(iv) High CMRR with high input impedance

(e) A 4-bit R-2R ladder DAC uses reference voltage of 8V. The analog output corresponding to digital input 1010 is

(i) 2.5V
(ii) 5V
(iii) 6V
(iv) 7.5V

(f) The free running frequency of a 555 timer in astable mode primarily depends upon

(i) External R-C network
(ii) Comparator offset voltage
(iii) Output transistor gain
(iv) Supply ripple factor

(g) The main function of a Phase Locked Loop IC 565 in communication systems is

(i) Voltage amplification
(ii) Frequency synthesis and synchronization
(iii) Current limiting
(iv) Rectification

(h) An LM317 regulator is adjusted using external resistors. If the adjustment terminal becomes open, the output voltage generally

(i) Becomes zero
(ii) Becomes fixed at 5V
(iii) Rises toward maximum regulated value
(iv) Oscillates at high frequency

(i) A second order active low pass filter has cut-off frequency 1kHz. If the input frequency is increased to 10kHz, the output magnitude will

(i) Attenuate significantly
(ii) Remain unchanged
(iii) Increase exponentially
(iv) Become unstable immediately

(j) The ICL8038 IC is widely used for generating

(i) PWM pulses only
(ii) RF carrier modulation only
(iii) Square, triangular and sine waves
(iv) Digital clock synchronization only

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Q.2 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the complete monolithic IC fabrication process including epitaxial growth, masking, etching and diffusion techniques.

(b) Discuss briefly operation of comparators, multivibrators and waveform generators using OP-AMPs, and design a Schmitt trigger circuit using $V_{sat} = \pm 12V$ such that the upper trigger point is 4V and lower trigger point is 2V. Determine suitable resistor values.

Q.3 **Solve both questions :** **[10+4=14]**

(a) Discuss the construction and fabrication of monolithic resistors, capacitors and FETs in integrated circuits, and determine the sheet resistance and capacitance value for a fabricated IC structure having resistive layer resistivity $\rho = 0.02\Omega m$, thickness 2µm, resistor dimensions 200µm×20µm, dielectric constant $\varepsilon_r = 3.9$, oxide thickness 0.1µm, and capacitor plate area 100µm×100µm.

(b) Explain the working of DAC and ADC circuits using operational amplifiers including weighted resistor.

Q.4 **Solve both questions :** **[7 x 2 = 14]**

(a) Describe ideal and practical characteristics of an operational amplifier including offset voltage, bias current, slew rate and CMRR, and calculate the closed loop gain and bandwidth of an amplifier having open loop gain 2×10^5 and unity gain bandwidth of 2 MHz.

(b) Explain the internal functional blocks and operating modes of IC 555 timer with suitable diagrams, and design an astable multivibrator using 555 timer to generate a 5 kHz square wave with duty cycle 60%.

Q.5 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the working principle of inverting and non-inverting amplifiers using OP-AMP with neat circuit diagrams and design an amplifier system that produces output $V_o = -5V_1 + 3V_2$ for given input voltages using suitable resistor selection.

(b) Describe the working principle and applications of PLL IC 565 and voltage controlled oscillator IC 566, and calculate the lock range and capture range of a PLL system having free running frequency 100 kHz, loop gain K=104, and supply voltage 12V.

Q.6 **Solve both questions :** **[7 x 2 = 14]**

(a) Discuss the operation and limitations of integrator and differentiator circuits using operational amplifiers, and design a practical integrator circuit for a square wave input of amplitude 4V and frequency 2 kHz to obtain a triangular wave output of peak amplitude 5V.

(b) Explain the operation of fixed and variable IC voltage regulators including LM78XX, LM79XX and LM317 with protection mechanisms, and design a regulated power supply producing variable output from 5V to 15V using LM317.

Q.7 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the construction and working of instrumentation amplifiers with emphasis on CMRR improvement and noise rejection, and design a three OP-AMP instrumentation amplifier having overall gain 100 using $R_1 = R_2 = 10k\Omega$. Determine the value of gain-setting resistor R_G.

(b) Discuss the architecture and applications of SMPS and LM380 power amplifier IC with suitable block diagrams, and calculate the efficiency, ripple factor and power dissipation of a switching regulator having input voltage 24V, output voltage 12V, output current 2A, and input power 30W.

Q.8 **Solve both questions :** **[7 x 2 = 14]**

(a) Describe first and second order active filters using operational amplifiers and compare Butterworth and Chebyshev responses, and design a second order low pass Butterworth filter with cut-off frequency 5 kHz, gain 2, and capacitances $C_1 = C_2 = 0.01\mu F$. Determine the required resistor values.

(b) Explain the differential amplifier configuration used in operational amplifiers with proper circuit analysis, and determine the differential gain, common mode gain and CMRR for a transistor differential pair having differential gain $A_d = 5000$ and common mode gain $A_c = 0.5$.

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Q.9 **Write short notes on any two of the following:** **[7 x 2 = 14]**

(a) Limitations of practical operational amplifiers in high frequency applications and methods for compensation

(b) Comparative analysis of weighted resistor DAC and R-2R ladder DAC in high resolution systems

(c) Design challenges in high precision active filters and instrumentation amplifiers for biomedical applications

(d) Role of PLL and function generator ICs in modern communication and embedded systems

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